Day 37 - Homework Session

Routhian Mechanics

https://en.wikipedia.org/wiki/Routhian_mechanics

Announcements

- Homework 8 due Friday, Nov 21st (late after Nov 26th)
- Next Week: Project Work and Discussion
- Last Week: Presentations
- Final Project Due Dec 8th (no later than 11:59 pm)
- **No Final Exam**

Complete Google Form

By November 21st

Reporting your group members for the final project and a short summary of your project idea for sharing with the class.

https://forms.gle/iPKR9EDAaHW3GirN7

Exercise 5 - Changing Coordinates

Consider two particles with masses $m_1 = m_2 = m$ connected by a spring with spring constant k. They sit on a frictionless surface but are constrained to the x-axis. The location of the first particle is x_1 and the location of the second particle is x_2. The spring is at equilibrium when the particles are a distance L apart. Assume the spring is massless and that the left mass never changes position with the right mass (i.e., the spring is always horizontal).

- 5a. Write down the kinetic and potential energy for this system in terms of x_1 and x_2 and their associated velocities.

In this problem we are changing the coordinates to explore a different aspect of the problem (and to make the math easier). This is a common technique that will not always be obvious. But practice will help us identify when it is useful.

Exercise 5 - Changing Coordinates

- 5b. Write down the Lagrangian for this system in terms of x_1 and x_2 and their associated velocities $\mathcal{L}(x_1, x_2, \dot{x}_1, \dot{x}_2)$. Find the 2 equations of motion.
- 5c. Now consider a new coordinate system where the center of mass is at $x_{cm} = \frac{1}{2}(x_1 + x_2)$ and the relative coordinate is $x = x_1 - x_2$. Write down the kinetic and potential energy in terms of x_{cm} and x and their associated velocities.

Exercise 5 - Changing Coordinates

- 5d. Write down the Lagrangian for this system in terms of the center of mass coordinate (x_{cm}) and the spring's extension (x) and the associated velocities $\mathcal{L}(x_{cm}, x, \dot{x}_{cm}, \dot{x})$. Find the 2 equations of motion.
- 5e. Find $x_{cm}(t)$ and $x(t)$ for the system for some generic initial conditions of your choosing. Describe the motion of the system.